Clinical Study Protocol

A Randomized, Double-Blind, Placebo-Controlled, Parallel-Group Study of Transdermal Xanomeline in Subjects With Mild to Moderate Alzheimer's Disease

Protocol CDISCPILOT01

May 18, 2026

Contents

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Field	Value
Protocol Number	CDISCPILOT01
Protocol Version	Version 1.0
Protocol Date	May 18, 2026
Investigational Product	Xanomeline transdermal patch
Sponsor	CDISCPILOT01 example sponsor
Development Phase	Clinical pharmacology, efficacy, and safety
Study Design	Randomized, double-blind, placebo-controlled, parallel-group
Confidentiality	This reconstructed protocol is for example-project documentation and data interpretation only.

1 Protocol Administrative Information

This protocol is reconstructed from the XPT source data in `inst/example-project/data/source/CDISCPILOT01`. It is intended for example-project documentation and data interpretation. It is not an original sponsor-approved protocol and must not be used to conduct human subject research.

Study identifier: CDISCPILOT01

Protocol title: A randomized, placebo-controlled, parallel-group study of transdermal Xanomeline in subjects with mild to moderate Alzheimer's disease.

Investigational product: Xanomeline transdermal patch.

Control: Matching placebo transdermal patch.

Country and sites: USA; 17 sites.

Observed subject reference period: 2012-07-09 to 2015-03-05.

Observed exposure period: 2012-07-09 to 2015-03-05.

1.1 Protocol History

Version	Date	Summary of Changes
1.0	2026-05-18	Initial reconstructed protocol prepared from available CDISCPILOT01 source domains.

2 Protocol Synopsis

Item	Description
Design	Multicenter, randomized, placebo-controlled, parallel-group clinical pharmacology and safety study.
Population	Adult and older adult subjects with Alzheimer's disease, supported by medical history records for 254 subjects.
Sample size in source data	306 subjects in demographics, including 254 non-screen-failure records and 254 records with non-screen-failure actual treatment.
Treatment groups	Placebo; Screen Failure; Xanomeline High Dose; Xanomeline Low Dose
Route and dosage form	Once-daily transdermal patch.
Primary objective	To determine if there is a statistically significant relationship between the change in both ADAS-Cog and CIBIC+ scores, and drug dose (0, 50 cm2 [54 mg], and 75 cm2 [81 mg]); To document the safety profile of the xanomeline TTS.
Key assessments	Adverse events, concomitant medications, disposition, ECG, exposure, laboratory tests, medical history, pharmacokinetic concentrations and parameters, subject visits, and vital signs.
PK sampling	Plasma and urine Xanomeline concentrations at baseline from pre-dose through 48 hours post-dose.

Item	Description
Key scheduled visits	SCREENING 1 (Day -7); SCREENING 2 (Day -1); BASELINE (Day 1); AMBUL ECG PLACEMENT (Day 13); WEEK 2 (Day 14); WEEK 4 (Day 28); AMBUL ECG REMOVAL (Day 30); WEEK 6 (Day 42); WEEK 8 (Day 56); WEEK 10 (T) (Day 70); WEEK 12 (Day 84); WEEK 14 (T) (Day 98); WEEK 16 (Day 112); WEEK 18 (T) (Day 126); WEEK 20 (Day 140); WEEK 22 (T) (Day 154); WEEK 24 (Day 168); WEEK 26 (Day 182); RETRIEVAL (Day 168)

3 Background and Rationale

Xanomeline is a cholinergic agonist evaluated here as a transdermal therapeutic system for subjects with mild to moderate Alzheimer's disease. The transdermal formulation is intended to provide systemic exposure while supporting once-daily administration.

The CDISCPILOT01 trial summary records describe a randomized, double-blind, placebo-controlled study evaluating dose response for ADAS-Cog and CIBIC+ scores, with additional safety, behavioral, activities-of-daily-living, and pharmacokinetic assessments. The reconstructed source package supports the protocol structure through trial summary, subject visit, adverse event, concomitant medication, disposition, ECG, exposure, laboratory, medical history, pharmacokinetic concentration, pharmacokinetic parameter, supplemental qualifier, and vital sign records.

The available source package still does not include the original sponsor protocol, randomization metadata, eligibility case report form fields, efficacy assessment domains, study drug accountability records, or detailed event narratives. Those omissions limit reconstruction of operational rules and final endpoint hierarchy. The protocol below uses the available source data as the controlling evidence and flags inferred content where appropriate.

4 Objectives and Endpoints

4.1 Primary Objective

To determine whether there is a statistically significant relationship between change in ADAS-Cog and CIBIC+ scores and Xanomeline transdermal dose.

4.2 Secondary Objectives

1. To document the safety profile of Xanomeline transdermal therapeutic system.
2. To assess dose-dependent improvement in behavior and activities of daily living.
3. To characterize plasma and urine Xanomeline concentration-time profiles and pharmacokinetic parameters after dosing.
4. To summarize laboratory, ECG, and vital sign findings over scheduled and unscheduled visits.
5. To summarize adverse events by term, seriousness, severity, treatment-emergent flag, and relationship to study drug.
6. To summarize disposition, concomitant medication, medical history, and exposure to placebo, Xanomeline 54 mg, and Xanomeline 81 mg transdermal patches.

4.3 Endpoints

Primary efficacy endpoints are inferred from trial summary records as change in ADAS-Cog and CIBIC+ scores by dose, but efficacy assessment domains are not present in this source package. Safety endpoints include adverse events, serious adverse events, adverse event severity, treatment-emergent flag and relationship, laboratory reference range indicators, ECG parameters, vital sign values, deaths recorded in demographics and disposition, concomitant medication use, and exposure interruptions or dosing patterns where available. Pharmacokinetic endpoints include observed Xanomeline concentration by matrix, nominal time point, visit, and derived PK parameters.

Treatment-emergent adverse event timing can be approximated from adverse event start dates and exposure dates. Detailed adjudication, causality narratives, and original event coding decisions would need the source clinical database or original study documents.

Table 1: Objectives and Endpoints

Objective	Endpoint
Primary efficacy	Change in ADAS-Cog and CIBIC+ scores by assigned dose.
Secondary efficacy	Behavior and activities of daily living endpoints, including NPI-X and DAD where available in the original efficacy database.
Safety and tolerability	Treatment-emergent adverse events, serious adverse events, laboratory tests, vital signs, ECGs, deaths, concomitant medications, and disposition.
Pharmacokinetics	Xanomeline plasma and urine concentrations and derived PK parameters including AUC, Cmax, Tmax, half-life, renal clearance, and urinary amount excreted.

5 Study Design

This is reconstructed as a multicenter, randomized, placebo-controlled, parallel-group study. Subjects were screened, randomized to placebo or one of two Xanomeline dose levels, and followed through Week 24 with additional Week 26 or retrieval records represented in vital signs. Exposure records show dosing at baseline, Week 2, and Week 24.

Table 2: Study Design

Attribute	Description
Study phase	Clinical pharmacology, efficacy, and safety study.
Design	Multicenter, randomized, double-blind, placebo-controlled, parallel-group study.
Control	PLACEBO
Blinding	DOUBLE BLIND
Population	Subjects with Mild to Moderate Alzheimer's Disease.
Planned duration	26 weeks
Treatment arms	Placebo; Xanomeline High Dose; Xanomeline Low Dose
Route and dosage form	Transdermal patch; QD; 12 to 14 hours transdermal application.
Study sites	17 sites in USA.

5.1 Study Schema

Subjects enter screening, complete baseline assessments, and are randomized to placebo, Xanomeline 54 mg, or Xanomeline 81 mg transdermal patch. Treatment and safety follow-up continue through Week 24, with Week 26 and retrieval records represented in the source data. PK sampling is concentrated at baseline from pre-dose through 48 hours post-dose.

Table 3: Planned Treatment Arms

Planned Arm	Subjects
Placebo	86
Screen Failure	52
Xanomeline High Dose	84
Xanomeline Low Dose	84

Table 4: Actual Treatment Arms

Actual Arm	Subjects
Placebo	86
Screen Failure	52
Xanomeline High Dose	72
Xanomeline Low Dose	96

Table 5: Subjects by Site

Site	Subjects
701	51
710	38
708	32
716	29
704	25
709	23
705	21
703	19
718	13
711	12
715	12
713	9
717	7
714	6
707	5
706	3
702	1

6 Study Population

The study population consists of adult and older adult subjects with mild to moderate Alzheimer's disease who are able to comply with study visits, treatment administration, and protocol-required safety and PK assessments. Eligibility criteria below are reconstructed from the available domains and should be treated as plausible protocol language rather than the original sponsor-approved criteria.

6.1 Inclusion Criteria

Based on the source data and the CDISC pilot Xanomeline context, eligible subjects would have met all of the following criteria:

1. Adult or older adult male or female subjects.
2. Clinical diagnosis of Alzheimer's disease, represented in the medical history domain for 254 subjects.
3. Ability to receive a transdermal patch and complete protocol-required safety and PK assessments.
4. Availability for follow-up through the planned 26 weeks treatment period.
5. Written informed consent provided by the subject or legally authorized representative.

6.2 Exclusion Criteria

Subjects would have been excluded for any of the following:

1. Clinically significant unstable medical condition that could confound safety assessment.
2. Screening laboratory or vital sign finding considered clinically unacceptable by the investigator.
3. Known hypersensitivity or contraindication to Xanomeline or patch components.
4. Use of prohibited concomitant medications or participation in another interventional study.
5. Any condition preventing reliable PK sampling or follow-up.

6.3 Withdrawal Criteria

Subjects may discontinue study treatment or withdraw from study participation because of adverse event, lack of efficacy, withdrawal by subject or caregiver, physician decision, protocol violation, death, loss to follow-up, or sponsor decision. These categories are represented in the disposition source data and should be handled according to the final statistical analysis plan.

6.4 Demographic Characteristics in Source Data

Table 6: Sex

Characteristic	Value
Subjects	306
Non-screen-failure planned treatment records	254
Non-screen-failure actual treatment records	254
Age, mean	75.1
Age, median	77
Age, range	50-89 years
Female	179
Male	127
Deaths flagged in demographics	3

Table 7: Race

Sex	Subjects
F	179
M	127

Table 8: Ethnicity

Race	Subjects
WHITE	273
BLACK OR AFRICAN AMERICAN	29
AMERICAN INDIAN OR ALASKA NATIVE	2
ASIAN	2

Ethnicity	Subjects
HISPANIC OR LATINO	17
NOT HISPANIC OR LATINO	289

7 Investigational Product and Administration

The investigational product is Xanomeline administered as a once-daily transdermal patch. Exposure records include placebo patches and active Xanomeline patches at 54 mg and 81 mg.

Study drug should be applied as a transdermal patch according to randomized treatment assignment. Subjects, caregivers, investigators, and sponsor personnel involved in study conduct are assumed to remain blinded to assigned treatment until the database is locked, unless unblinding is required for subject safety.

Table 9: Exposure Regimens

	Treatment	Dose	Unit	Form	Frequency	Route
1	PLACEBO	0	mg	PATCH	QD	TRANSDERMAL
6	XANOMELINE	54	mg	PATCH	QD	TRANSDERMAL
7	XANOMELINE	81	mg	PATCH	QD	TRANSDERMAL

Table 10: Exposure Visits

Visit Number	Visit	Study Day
3	BASELINE	1
4	WEEK 2	14
12	WEEK 24	168

7.1 Treatment Compliance and Accountability

Treatment compliance, patch application site rotation, accountability, and dose modification rules are not represented in the source domains and would need to be supplied by the original protocol or operational manuals. In a sponsor protocol, accountability would include dispensation, return, reconciliation, and documentation of missed or interrupted patch applications.

7.2 Prior and Concomitant Therapy

Concomitant medications should be recorded from screening through the final study visit. Medications used for the primary study condition, prophylaxis, or treatment of adverse events should be coded and summarized by treatment arm.

8 Schedule of Assessments

The following schedule is reconstructed from visit, laboratory, vital sign, ECG, exposure, PK, disposition, and subject-visit records. Unscheduled visits are retained because the source data include clinically relevant safety observations outside nominal scheduled visits.

Table 11: Visits Represented in Source Data

	Visit Number	Visit	Study Day
1	1.0	SCREENING 1	-7
39317	1.1	UNSCHEDULED 1.1	NA
100843	1.2	UNSCHEDULED 1.2	NA
45196	1.2	UNSCHEDULED 1.2	NA
100844	1.3	UNSCHEDULED 1.3	NA

	Visit Number	Visit	Study Day
45078	1.3	UNSCHEDULED 1.3	NA
2	2.0	SCREENING 2	-1
3	3.0	BASELINE	1
103124	3.1	UNSCHEDULED 3.1	NA
7531	3.5	AMBUL ECG PLACEMENT	13
4	4.0	WEEK 2	14
40955	4.1	UNSCHEDULED 4.1	NA
54234	4.2	UNSCHEDULED 4.2	NA
101407	4.2	UNSCHEDULED 4.2	NA
5	5.0	WEEK 4	28
35912	5.1	UNSCHEDULED 5.1	NA
6	6.0	AMBUL ECG REMOVAL	30
35386	6.1	UNSCHEDULED 6.1	NA
7	7.0	WEEK 6	42
71577	7.1	UNSCHEDULED 7.1	NA
8	8.0	WEEK 8	56
100232	8.1	WEEK 10 (T)	70
55921	8.2	UNSCHEDULED 8.2	NA
9	9.0	WEEK 12	84
100208	9.1	WEEK 14 (T)	98
102752	9.1	UNSCHEDULED 9.1	NA
37612	9.2	UNSCHEDULED 9.2	NA
37613	9.3	UNSCHEDULED 9.3	NA
100406	9.3	UNSCHEDULED 9.3	NA
10	10.0	WEEK 16	112
100236	10.1	WEEK 18 (T)	126
100663	10.2	UNSCHEDULED 10.2	NA
11	11.0	WEEK 20	140
100211	11.1	WEEK 22 (T)	154
101010	11.2	UNSCHEDULED 11.2	NA
12	12.0	WEEK 24	168
62453	12.1	UNSCHEDULED 12.1	NA
13	13.0	WEEK 26	182
90484	13.1	UNSCHEDULED 13.1	NA
72	101.0	AE FOLLOW-UP	NA
73	201.0	RETRIEVAL	168

Table 12: Reconstructed Schedule of Assessments

Assessment	Screening	Baseline	Weeks 2-24	Week 26/Retrieval	Unscheduled
Demographics	X				
Medical history	X				
Adverse events		X	X		X
Concomitant medications		X	X		X
Disposition	X	X	X	X	
ECG	X	X	X	X	
Exposure		X	X		
Laboratory tests	X	X	X		X
Pharmacokinetic concentrations		X			
Pharmacokinetic parameters		X	X		
Subject visits	X	X	X	X	
Vital signs	X	X	X	X	X

9 Safety Assessments

Safety will be monitored throughout the study by review of adverse events, concomitant medications, clinical laboratory tests, vital signs, ECGs, deaths, and disposition events. Safety data should be reviewed by visit and treatment arm, with clinically significant findings followed until resolution or stabilization where possible.

9.1 Adverse Events

Adverse events should be collected from first study treatment through the end of follow-up and coded using a standard medical dictionary. Events should be classified by seriousness, severity, relationship to study drug, action taken, outcome, and treatment-emergent status. The adverse event source domain contains 1191 records for 225 subjects. There are 3 serious adverse event records across 3 subjects and 43 severe adverse event records across 31 subjects.

Table 13: Most Frequent Adverse Events

Adverse Event	Records
PRURITUS	84
APPLICATION SITE PRURITUS	78
ERYTHEMA	59
APPLICATION SITE ERYTHEMA	46
RASH	45
APPLICATION SITE IRRITATION	41
APPLICATION SITE DERMATITIS	36
DIZZINESS	34
SKIN IRRITATION	25
SINUS BRADYCARDIA	24
DIARRHOEA	21
HEADACHE	21
NASOPHARYNGITIS	21
NAUSEA	21
COUGH	20

Table 14: Adverse Events by Severity and Seriousness

	Severity	Serious	Records
1	MILD	N	770
2	MODERATE	N	377
3	SEVERE	N	41
5	MODERATE	Y	1
6	SEVERE	Y	2

Table 15: Adverse Event Relationship to Study Drug

Relationship to Study Drug	Records
PROBABLE	361
POSSIBLE	343
NONE	322
REMOTE	161
Not reported	4

Table 16: Treatment-Emergent Adverse Event Flag

Treatment-Emergent Flag	Records
Y	1126
N	65

9.2 Laboratory Assessments

Clinical laboratory testing includes chemistry, hematology, urinalysis, and additional miscellaneous results. Abnormal findings should be assessed for clinical significance and followed according to investigator judgment. The source package contains 59580 laboratory records.

Table 17: Laboratory Results by Category

Category	Results
CHEMISTRY	32740
HEMATOLOGY	21919
URINALYSIS	4370
OTHER	543
Not reported	8

Table 18: Laboratory Results by Reference Range Indicator

Reference Range Indicator	Results
NORMAL	56855
HIGH	1538
LOW	864
ABNORMAL	318
Not reported	5

Table 19: Laboratory Tests

	Category	Code	Test	Unit
1	CHEMISTRY	ALB	Albumin	g/L
11	CHEMISTRY	ALP	Alkaline Phosphatase	U/L
21	CHEMISTRY	ALT	Alanine Aminotransferase	U/L
32	CHEMISTRY	AST	Aspartate Aminotransferase	U/L
50	CHEMISTRY	BILI	Bilirubin	umol/L
60	CHEMISTRY	BUN	Blood Urea Nitrogen	mmol/L
70	CHEMISTRY	CA	Calcium	mmol/L
80	CHEMISTRY	CHOL	Cholesterol	mmol/L
90	CHEMISTRY	CK	Creatine Kinase	U/L
100	CHEMISTRY	CL	Chloride	mmol/L
114	CHEMISTRY	CREAT	Creatinine	umol/L
132	CHEMISTRY	GGT	Gamma Glutamyl Transferase	U/L
142	CHEMISTRY	GLUC	Glucose	mmol/L
172	CHEMISTRY	K	Potassium	mmol/L
246	CHEMISTRY	PHOS	Phosphate	mmol/L
266	CHEMISTRY	PROT	Protein	g/L
232	CHEMISTRY	SODIUM	Sodium	mmol/L

	Category	Code	Test	Unit
291	CHEMISTRY	URATE	Urate	umol/L
31	HEMATOLOGY	ANISO	Anisocytes	Not reported
42	HEMATOLOGY	BASO	Basophils	GI/L
59533	HEMATOLOGY	BASOLE	Basophils/Leukocytes	FRACTION
124	HEMATOLOGY	EOS	Eosinophils	GI/L
59535	HEMATOLOGY	EOSLE	Eosinophils/Leukocytes	FRACTION
152	HEMATOLOGY	HCT	Hematocrit	1
162	HEMATOLOGY	HGB	Hemoglobin	mmol/L
186	HEMATOLOGY	LYM	Lymphocytes	GI/L
59537	HEMATOLOGY	LYMLE	Lymphocytes/Leukocytes	FRACTION
609	HEMATOLOGY	MACROCY	Macrocytes	Not reported
194	HEMATOLOGY	MCH	Ery. Mean Corpuscular Hemoglobin	fmol(Fe)
204	HEMATOLOGY	MCHC	Ery. Mean Corpuscular HGB Concentration	mmol/L
214	HEMATOLOGY	MCV	Ery. Mean Corpuscular Volume	fL
25555	HEMATOLOGY	MICROCY	Microcytes	Not reported
224	HEMATOLOGY	MONO	Monocytes	GI/L
59539	HEMATOLOGY	MONOLE	Monocytes/Leukocytes	FRACTION
256	HEMATOLOGY	PLAT	Platelet	GI/L
19390	HEMATOLOGY	POIKILO	Poikilocytes	Not reported
684	HEMATOLOGY	POLYCHR	Polychromasia	Not reported
276	HEMATOLOGY	RBC	Erythrocytes	TI/L
306	HEMATOLOGY	WBC	Leukocytes	GI/L
15200	Not reported	HBA1C	Hemoglobin A1C	1
290	OTHER	TSH	Thyrotropin	mU/L
305	OTHER	VITB12	Vitamin B12	pmol/L
110	URINALYSIS	COLOR	Color	Not reported
182	URINALYSIS	KETONES	Ketones	Not reported
242	URINALYSIS	PH	pH	Not reported
286	URINALYSIS	SPGRAV	Specific Gravity	Not reported
301	URINALYSIS	UROBIL	Urobilinogen	Not reported

9.3 Vital Signs

Vital signs include blood pressure, pulse rate, height, temperature, and weight. Blood pressure and pulse should be collected in supine and standing positions where applicable.

Table 20: Vital Sign Parameters

	Code	Vital Sign	Position	Location	Unit
2	DIABP	Diastolic Blood Pressure	STANDING	Not reported	mmHg
4965	DIABP	Diastolic Blood Pressure	STANDING	Not reported	Not reported
1	DIABP	Diastolic Blood Pressure	SUPINE	Not reported	mmHg
43	HEIGHT	Height	Not reported	Not reported	cm
45	PULSE	Pulse Rate	STANDING	Not reported	BEATS/MIN
4996	PULSE	Pulse Rate	STANDING	Not reported	Not reported
44	PULSE	Pulse Rate	SUPINE	Not reported	BEATS/MIN
21787	PULSE	Pulse Rate	SUPINE	Not reported	Not reported
87	SYSBP	Systolic Blood Pressure	STANDING	Not reported	mmHg
5026	SYSBP	Systolic Blood Pressure	STANDING	Not reported	Not reported
86	SYSBP	Systolic Blood Pressure	SUPINE	Not reported	mmHg
21811	SYSBP	Systolic Blood Pressure	SUPINE	Not reported	Not reported
12138	TEMP	Temperature	Not reported	EAR	C
128	TEMP	Temperature	Not reported	ORAL CAVITY	C

	Code	Vital Sign	Position	Location	Unit
142	WEIGHT	Weight	Not reported	Not reported	kg

9.4 ECG Assessments

Standard and ambulatory ECG assessments should be performed according to the schedule of assessments. ECGs should be interpreted by qualified personnel, and clinically significant abnormalities should be recorded as adverse events where appropriate. The ECG domain contains 26717 records for 254 subjects.

Table 21: ECG Parameters

	Code	ECG Parameter	Unit
1	ECGINT	ECG Interpretation	Not reported
12	HR	Heart Rate	beats/min
54	QT	QT Duration	ms
96	RR	RR Duration	ms

9.5 Deaths

The demographics domain contains 3 subjects flagged with death. The disposition domain contains 3 subjects with a death disposition event. Adverse event data identify 3 subjects with serious adverse events. Detailed causality narratives and adjudication records are not represented in these source files.

10 Disposition, Medication, and Medical History

10.1 Disposition

Subject disposition should be summarized for all screened subjects and by randomized treatment arm. The disposition domain contains 850 records for 306 subjects. It includes randomization records, completion status, screen failures, discontinuations, and final visit milestones.

Table 22: Disposition Events

Disposition Event	Records
FINAL LAB VISIT	254
RANDOMIZED	254
COMPLETED	110
ADVERSE EVENT	92
SCREEN FAILURE	52
FINAL RETRIEVAL VISIT	36
WITHDRAWAL BY SUBJECT	27
STUDY TERMINATED BY SPONSOR	7
PROTOCOL VIOLATION	6
LACK OF EFFICACY	4
DEATH	3
PHYSICIAN DECISION	3
LOST TO FOLLOW-UP	2

Table 23: Supplemental Disposition Qualifiers

Qualifier	Label	Value
ENTCRIT	PROTOCOL ENTRY CRITERIA NOT MET	16
ENTCRIT	PROTOCOL ENTRY CRITERIA NOT MET	25

10.2 Concomitant Medications

The concomitant medication domain contains 7510 records for 229 subjects.

Table 24: Concomitant Medication Indications

Indication	Records
Not reported	4173
PROPHYLAXIS OR NON-THERAPEUTIC USE	3237
PRIMARY STUDY CONDITION	100

Table 25: Most Frequent Concomitant Medications

Medication	Records
UNCODED	6085
ACETYLSALICYLIC ACID	380
ESTROGENS CONJUGATED	195
CALCIUM	180
AMLODIPINE	112
HYDROCORTISONE	105
NIZATIDINE	53
FUROSEMIDE	48
NAPROXEN SODIUM	46
DIGOXIN	39
ALGELDRATE	36
DOXAZOSIN MESILATE	32
METFORMIN HYDROCHLORIDE	23
FLUVASTATIN	21
SALBUTAMOL SULFATE	21

10.3 Medical History

The medical history domain contains 1818 records for 254 subjects, including Alzheimer's disease records for 254 subjects.

Table 26: Medical History Categories

Medical History Category	Records
SIGNIFICANT PRE-EXISTING CONDITION	858
HISTORICAL DIAGNOSIS	706
PRIMARY DIAGNOSIS	254

Table 27: Most Frequent Medical History Terms

Medical History Term	Records
ALZHEIMER'S DISEASE	254
VERBATIM_0342	58
VERBATIM_1309	39
VERBATIM_0936	30
VERBATIM_1775	23
VERBATIM_1716	21
VERBATIM_0443	19
VERBATIM_0308	17
VERBATIM_1512	17
VERBATIM_0164	14
VERBATIM_0301	13
VERBATIM_0391	13
VERBATIM_0929	12
VERBATIM_1769	12
VERBATIM_0809	11

11 Pharmacokinetic Assessments

The pharmacokinetic concentration domain contains 4572 records for Xanomeline in plasma and urine. Sampling is represented at baseline from pre-dose through 48 hours post-dose, including interval collections for urine. The pharmacokinetic parameter domain contains 2688 records for 168 subjects.

Table 28: PK Analytes and Specimens

	Code	Analyte	Unit	Specimen	Lower Limit
1	XAN	XANOMELINE	ug/ml	PLASMA	0.01
15	XAN	XANOMELINE	ug/ml	URINE	0.01

Table 29: PK Nominal Time Points

	Visit	Study Day	Nominal Time Point	Nominal Hour
1	BASELINE	1	Pre-dose	-0.50
2	BASELINE	1	5 Min Post-dose	0.08
3	BASELINE	1	30 Min Post-dose	0.50
4	BASELINE	1	1h Post-dose	1.00
5	BASELINE	1	1.5h Post-dose	1.50
6	BASELINE	1	2h Post-dose	2.00
15	BASELINE	1	0-6h Post-dose	3.00
7	BASELINE	1	4h Post-dose	4.00
8	BASELINE	1	6h Post-dose	6.00
9	BASELINE	1	8h Post-dose	8.00
16	BASELINE	1	6-12h Post-dose	9.00
10	BASELINE	1	12h Post-dose	12.00
11	BASELINE	1	16h Post-dose	16.00
17	BASELINE	1	12-24h Post-dose	18.00
12	BASELINE	1	24h Post-dose	24.00
13	BASELINE	1	36h Post-dose	36.00
18	BASELINE	1	24-48h Post-dose	37.00
14	BASELINE	1	48h Post-dose	48.00

Visit	Study Day	Nominal Time Point	Nominal Hour

Table 30: PK Parameters

	Code	Parameter	Unit	Specimen
1	AUCALL	AUC All	h*ug/ml	PLASMA
5	AUCLST	AUC to Last Nonzero Conc	h*ug/ml	PLASMA
6	CLST	Last Nonzero Conc	ug/ml	PLASMA
10	CMAX	Max Conc	ug/ml	PLASMA
11	LAMZ	Lambda z	/h	PLASMA
12	LAMZHL	Half-Life Lambda z	h	PLASMA
13	LAMZNPT	Number of Points for Lambda z	U	PLASMA
16	TMAX	Time of CMAX	h	PLASMA
14	RCAMINT	Ae	mg	URINE
15	RENALCL	CLR	mL/min	URINE

PK analyses should summarize concentration-time data by treatment, specimen, and nominal time point. Derived PK parameters should be summarized by specimen, treatment arm, and parameter. Values below the lower limit of quantification should be handled using predefined rules before noncompartmental analysis or descriptive summaries.

12 Statistical Considerations

12.1 Analysis Populations

All analyses for this reconstructed protocol should be descriptive unless the original efficacy analysis plan is supplied. The safety analysis set should include all subjects with non-screen-failure actual treatment assignment, using the demographics actual arm field and supplemental population flags as the available proxy for treatment received. The intent-to-treat and efficacy populations are represented by supplemental demographic flags but cannot be fully audited without the original randomization and efficacy datasets. PK summaries should include subjects with at least one quantifiable or reportable PK concentration or parameter record.

Table 31: Supplemental Population Flags

Flag	Population	Value
COMPLT16	Completers of Week 16 Population Flag	Y
COMPLT24	Completers of Week 24 Population Flag	Y
COMPLT8	Completers of Week 8 Population Flag	Y
EFFICACY	Efficacy Population Flag	Y
ITT	Intent to Treat Population Flag	Y
SAFETY	Safety Population Flag	Y

12.2 General Analysis Methods

Continuous variables should be summarized with n, mean, standard deviation, median, minimum, and maximum. Categorical variables should be summarized with counts and percentages. Adverse events should be summarized by system organ class, preferred term, severity, seriousness, treatment-emergent flag, relationship, and treatment arm. Disposition, concomitant medication, medical history, ECG, laboratory, and vital sign summaries should be presented by visit or timing category and treatment arm where applicable. PK concentrations and parameters should be summarized by specimen, nominal time point or parameter, and actual treatment arm.

12.3 Efficacy Analysis

The trial summary records identify dose-response evaluation of ADAS-Cog and CIBIC+ as the primary objective. Because efficacy assessment domains are not present in the example source package, the efficacy analysis cannot be reconstructed here. A complete protocol would specify estimands, analysis windows, missing data handling, covariates, multiplicity control, and sensitivity analyses.

12.4 Safety Analysis

Safety analyses should include all treated subjects and summarize observed results by actual treatment arm. Treatment-emergent adverse events should be summarized separately from all adverse events using the supplemental AE treatment-emergent flag.

13 Data Handling and Quality Control

The source package should be checked for:

1. Consistency of `STUDYID`, `USUBJID`, site, arm, and actual arm values across domains.
2. Consistency of visit numbers, visit names, and study days.
3. Consistency of trial summary records with the reconstructed protocol text.
4. Completeness of subject visit and exposure records for treated subjects.
5. Consistency of adverse event timing, seriousness, severity, treatment-emergent flag, relationship, and disposition outcomes.
6. Completeness and plausibility of concomitant medication and medical history records.
7. Validity of ECG, laboratory, vital sign, and PK standardized units.
8. Consistency of PK time point labels, nominal time values, and derived parameter records.
9. Traceability from generated outputs to the source XPT and supplemental qualifier records.

14 Ethical and Regulatory Considerations

For an actual trial, the protocol and informed consent materials would require institutional review board or ethics committee approval before initiation. The study would be conducted in accordance with Good Clinical Practice, applicable regulations, and the Declaration of Helsinki.

15 References

1. International Council for Harmonisation. ICH E6(R2): Good Clinical Practice.
2. International Council for Harmonisation. ICH E9: Statistical Principles for Clinical Trials.
3. CDISC Study Data Tabulation Model implementation principles, as reflected in the CDISCPILOT01 source domains.

16 Limitations of Reconstruction

This document is inferred from available XPT source data. Trial summary records provide objective, design, indication, and treatment-duration evidence, but the source package does not include the original protocol, efficacy assessment domains, randomization metadata, detailed adverse event narratives, adjudication records, or study drug accountability details. Those records should be incorporated from source systems or original study documents before using this as anything beyond an example protocol.

17 Appendix: Source Data Evidence

The following tables document the source domains and trial summary records used to reconstruct this example protocol.

Table 32: Source Domains Used for Reconstruction

Domain	Description	Records	Subjects
DM	Demographics	306	306
AE	Adverse events	1191	225
CM	Concomitant medications	7510	229
DS	Disposition	850	306
EG	ECG test results	26717	254
EX	Exposure	591	254
LB	Laboratory tests	59580	254
MH	Medical history	1818	254
PC	Pharmacokinetic concentrations	4572	254
PP	Pharmacokinetic parameters	2688	168
SUPPAE	Supplemental adverse event qualifiers	1191	225
SUPPDM	Supplemental demographic qualifiers	1197	254
SUPPDS	Supplemental disposition qualifiers	3	3
SV	Subject visits	3559	306
TS	Trial summary	33	NA
VS	Vital signs	29643	254

Table 33: Trial Summary Records

Item	Value
Trial indication	Mild to Moderate Alzheimer's Disease
Diagnosis group	Patients with Probable Mild to Moderate Alzheimer's Disease
Age range	50 years
Blinding	DOUBLE BLIND
Control type	PLACEBO
Treatment length	26 weeks
Dosing frequency	QD; 12 to 14 hours transdermal application
Primary objective	To determine if there is a statistically significant relationship between the change in both ADAS-Cog and CIBIC+ scores, and drug dose (0, 50 cm2 [54 mg], and 75 cm2 [81 mg]); To document the safety profile of the xanomeline TTS.

Item	Value
Secondary objectives	To assess the dose-dependent improvement in behavior. Improved scores on the Revised Neuropsychiatric Inventory (NPI-X) will indicate improvement in these areas.; To assess the dose-dependent improvements in activities of daily living. Improved scores on the Disability Assessment for Dementia (DAD) will indicate improvement in these areas.; To assess the dose-dependent improvements in an extended assessment of cognition that integrates attention/concentration tasks. The ADAS-Cog (14) will be used for this assessment.; To assess the treatment response as a function of Apo E genotype.